

Sharayu Sunil Mhaske

Angular Developer | **FullStack Developer**

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ABOUT ME

Software Developer with professional experience in building **Angular** and **TypeScript** web applications and developing backend services using **Java** and **Spring Boot** at **Fraunhofer FKIE**. Skilled in **REST APIs**, **RxJS**, state management, and software testing. Currently completing a **Master's in Web and Data Science** at **Universität Koblenz**, with additional experience in **Python**, **Machine Learning**, and **NLP tasks**. Seeking a **Full-Stack Software Developer** role.

WORK EXPERIENCE

Student Assistant — Fraunhofer FKIE

01/05/2024 – Current

Bonn, Germany

- Developed a dynamic hierarchical tree component using **Angular**, **TypeScript**, and recursive functions to generate parent-child nodes from custom **JSON** data.
- Implemented drag-and-drop functionality using **Angular CDK**, dynamically connecting drop lists to enable node reordering and resolve overlapping drop zones.
- Built **Angular Reactive Forms** using **FormBuilder** and **FormGroup** to create three node types, and managed real-time tree state synchronization using **RxJS BehaviorSubject**.
- Developed **RESTful APIs** using **Java** and **Spring Boot** to traverse, retrieve, update, and synchronize hierarchical tree data between the Angular frontend and backend using **HTTP** and **JSON**.
- Reconstructed a multi-component implementation into a single reusable **Angular** component, reducing component complexity by approximately **80%** while maintaining dynamic tree functionality.

Master's Thesis — Fraunhofer FKIE

01/05/2026 – Current

Bonn, Germany

- Fine-tuned and evaluated transformer-based models, **RoBERTa** and **DistilBERT**, for emotion intensity detection using the **BRIGHTER dataset**.
- Designed and compared **single-step** and **hierarchical two-step classification approaches**, with **Hierarchical RoBERTa** achieving the best overall performance.
- Conducted **5-fold cross-validation**, **error analysis**, and **LIME-based model interpretation** to evaluate model reliability, understand predictions, and identify challenges in emotion intensity detection.

Software Development Intern — Star Spring Works Pvt Ltd

01/09/2021 – 01/03/2022

Mumbai, India

- Developed and maintained multiple **Angular** and **TypeScript** applications for **B2C** clients, implementing reusable components and application features.
- Applied the **Facade Design Pattern** to centralize and abstract **API service calls**, improving code modularity, maintainability, and separation of concerns.
- Integrated device camera functionality to capture and upload images, handling **Blob data** and integrating **Firebase Firestore** using **AngularFire**.
- Implemented custom reactive state management using **RxJS** and **BehaviorSubject** to manage application data and synchronize state across components.
- Developed and maintained **unit tests** using **Jasmine** and **Karma**, supporting automated testing and **CI/CD** workflows, while managing source code with **Git** and **GitHub**.

SKILLS

Java, Python, JavaScript, TypeScript, HTML5, CSS3, SCSS, Angular, Angular Material, Angular CDK, Angular Reactive Forms, RxJS, Node.js, Spring Boot, RESTful APIs, MySQL, Firebase Firestore, Docker, Git, GitHub, Docker, CI/CD, Jasmine, Karma, Unit Testing

PROJECTS

FairRecruit AI - Full-Stack Recruitment Analytics Platform

2026 - Present

GitHub Project

- Extended the published research paper **"Bias in Recruitment Systems Utilizing LLMs" (ICAAI 2025)** into a full-stack application for recruitment analytics and AI fairness evaluation.
- Developed a modular application using **Angular, TypeScript, Python, and FastAPI**, incorporating **REST APIs** and frontend-backend integration.
- Established the foundation for a **Dataset Explorer** and bias analysis platform, using **Docker, GitHub Actions (CI/CD), and automated testing** to support maintainability and scalability.

Bias in Recruitment Systems Utilizing LLMs

01/10/2024 – 28/02/2025

Koblenz University

- Investigated **gender, racial, and geographical biases** in AI-driven recruitment using an adapted **Word Embedding Association Test (WEAT)**.
- Analyzed and visualized bias scores to identify demographic disparities; co-authored research published in the **ICAAI 2025 proceedings**.

EDUCATION

Master's in Web and Data Science

10/2023 – Current

University of Koblenz, Germany

Bachelor's in Computer Engineering

17/06/2017 – 15/05/2021

University of Mumbai, India

Final Grade: 9.53 (German Equivalent: 1.24)

LANGUAGES

German (B1) | English (Fluent) | Hindi (Fluent) | Marathi (Native)

ACHIEVEMENTS

Published Research Paper — Bias in Recruitment Systems Utilizing LLMs, ICAAI 2025
dl.acm.org/doi/10.1145/3787279.3787300

First Prize — Technical Project Presentation at XTACY 2k18

GATE Qualified — 2022